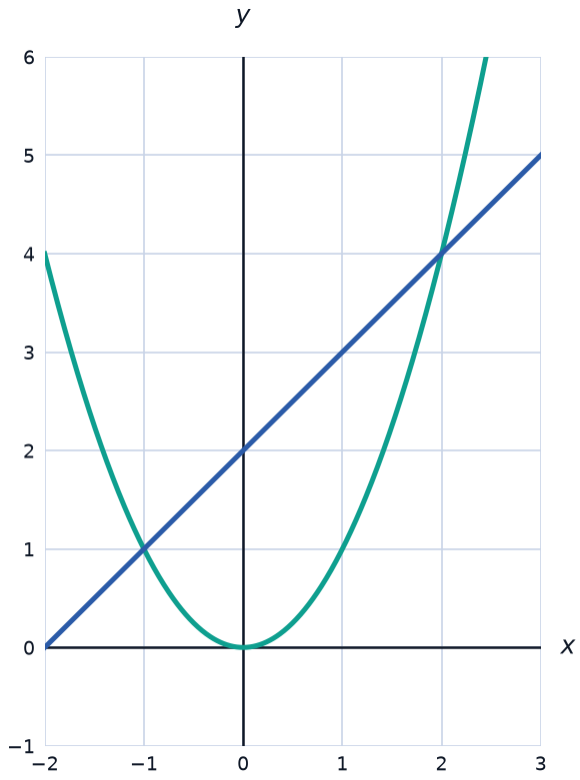
Area Between Curves

ClipSAT Printable Practice Suite (clipsat.org) • Topic 8.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

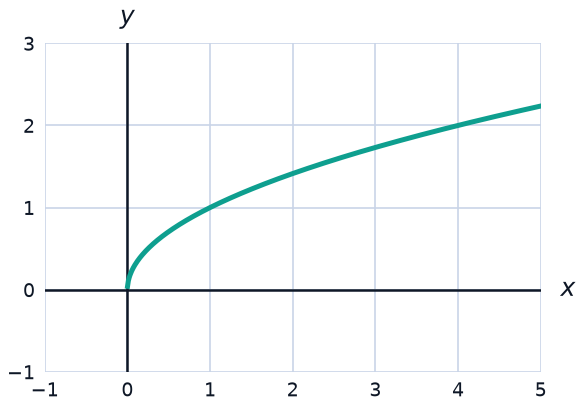
1 [Foundational] • Target: Area between two curves

Find the area of the region bounded by $y = x^2$ and $y = x + 2$.



2 [Foundational] • Target: Area between two curves

The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.



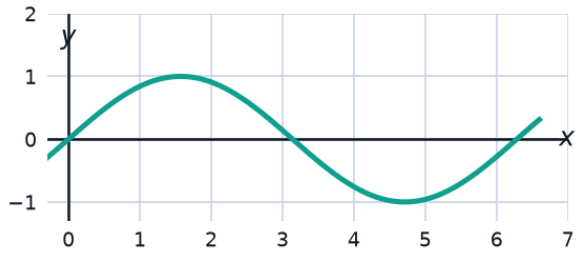
3 [Foundational] • Target: Setting up with respect to y, and total area

Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y.

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4 [Foundational] • Target: Setting up with respect to y, and total area

Find the total area between $f(x) = \sin x$ and the x-axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).

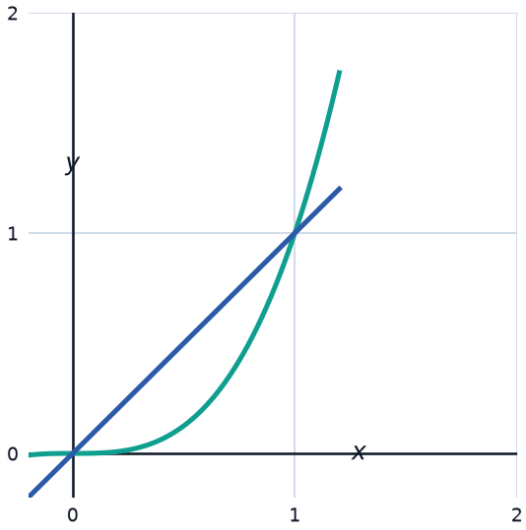


5 [Foundational] • Target: More area problems

Find the area of the region bounded by $y = x^2 - 4$ and $y = 4 - x^2$.

6 [Foundational] • Target: More area problems

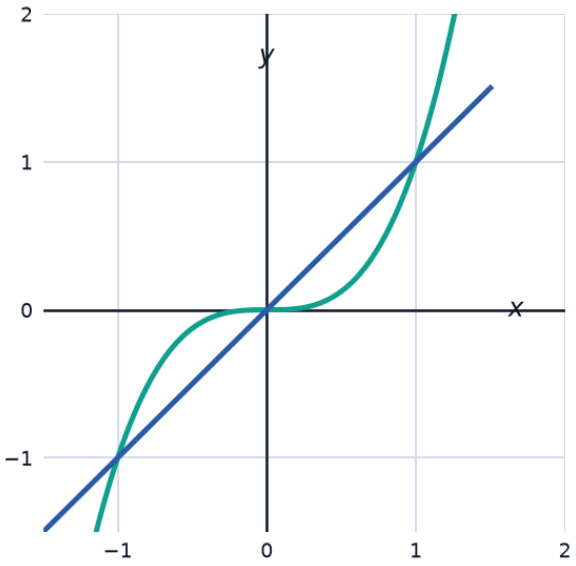
Find the area between $y = x^3$ and $y = x$ on $[0, 1]$ (note that $y = x$ is above $y = x^3$ on this interval).



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7 [Foundational] • Target: Areas requiring case splits

Find the area between $y = x^3$ and $y = x$ on $[-1, 1]$ (note the curves cross at $x = 0$, so the identity of the upper curve switches).

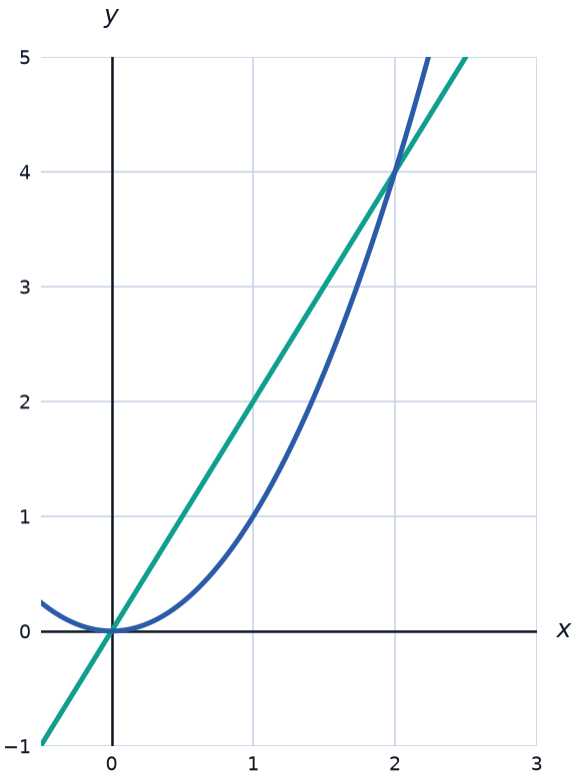


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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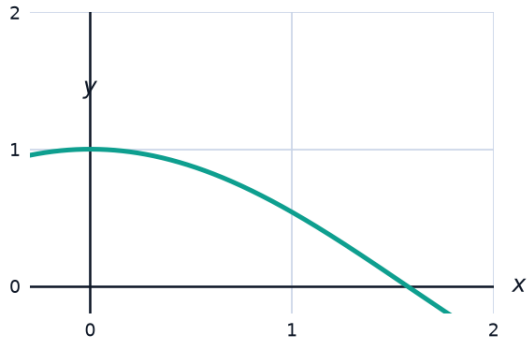
8 [Intermediate] • Target: Areas requiring case splits

Find the area enclosed between $y = 2x$ and $y = x^2$ on the interval where $y = 2x$ is above $y = x^2$.



9 [Intermediate] • Target: Areas involving trigonometric and exponential curves

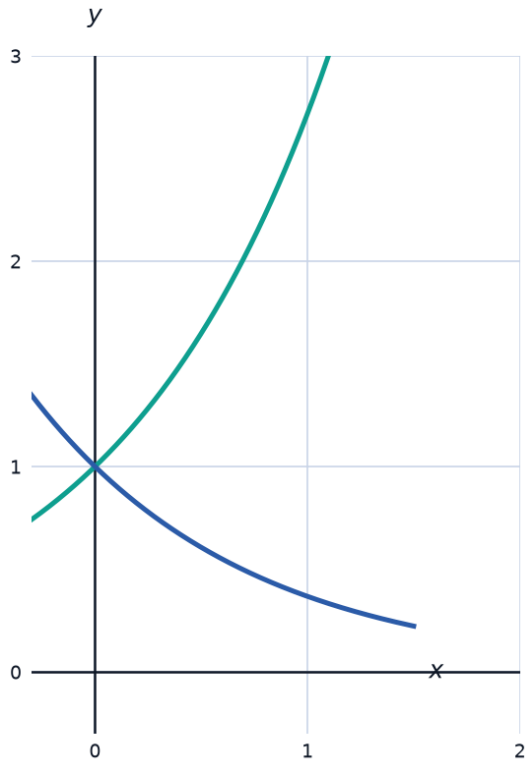
Find the area between $y = \cos x$ and $y = 0$ on $[0, \frac{\pi}{2}]$.



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10 **[Intermediate]** • **Target: Areas involving trigonometric and exponential curves**

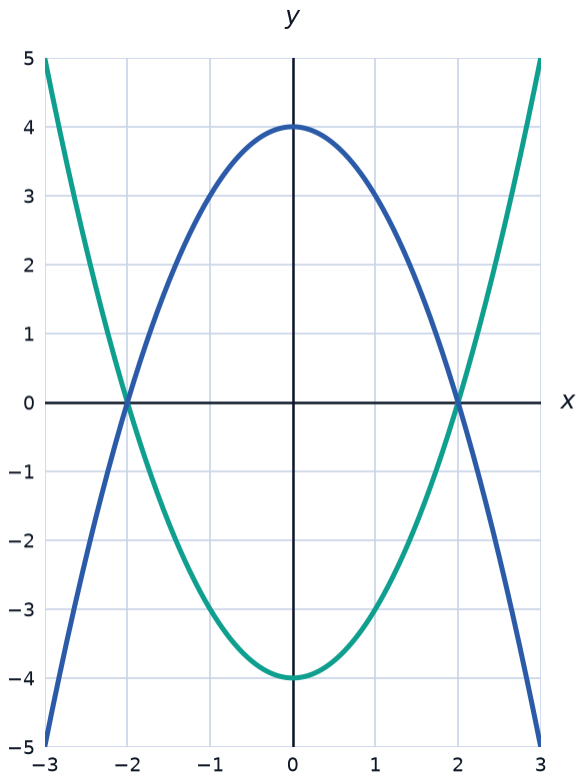
Find the area between $y = e^{x}$ and $y = e^{-x}$ on $[0, 1]$ (note $e^{x} \geq e^{-x}$ on this interval).



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11 **[Intermediate]** • **Target: Visualizing a region bounded by two curves**

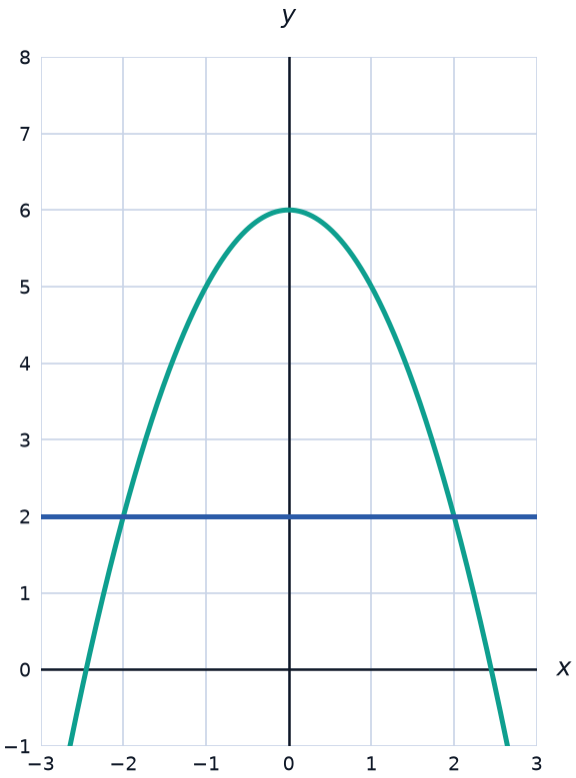
The graphs of $f(x) = x^2 - 4$ and $g(x) = 4 - x^2$ from an earlier problem are shown together. Confirm from the picture that the region is symmetric about the y-axis, and use this to verify the earlier area answer $\frac{64}{3}$ using $2\int_0^2 (8 - 2x^2)\,dx$.



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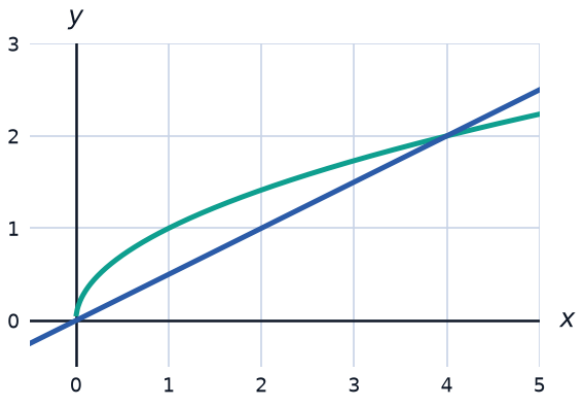
12 [Intermediate] • Target: Area between a curve and a horizontal line

Find the area between $y = 6 - x^2$ and $y = 2$.



13 [Intermediate] • Target: Area between a curve and a horizontal line

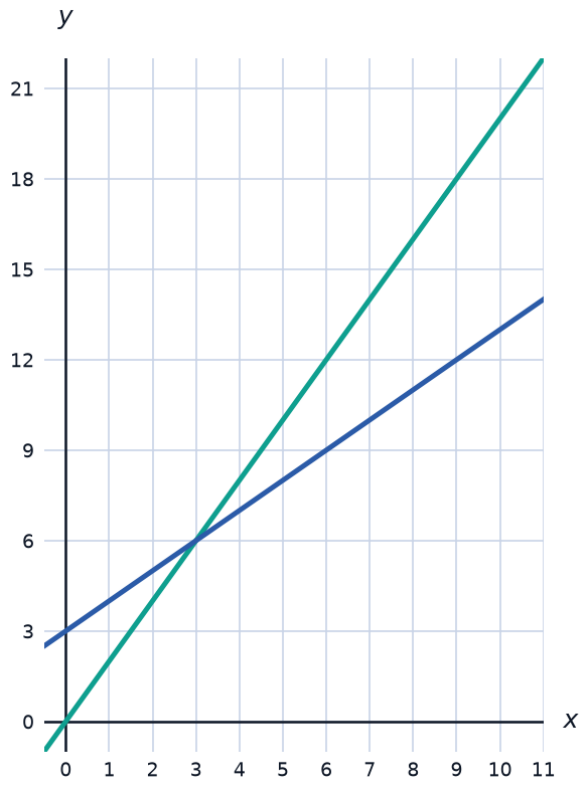
Find the area between $y = \sqrt{x}$ and $y = \frac{x}{2}$ for $x \geq 0$.



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14 **[Intermediate]** • **Target: Area between curves in applied contexts**

Two runners' velocities (m/s) over a 10-second race are $v_1(t) = 2t$ and $v_2(t) = t + 3$. The area between these curves represents the gap in distance traveled. Find this gap over $[0, 10]$.

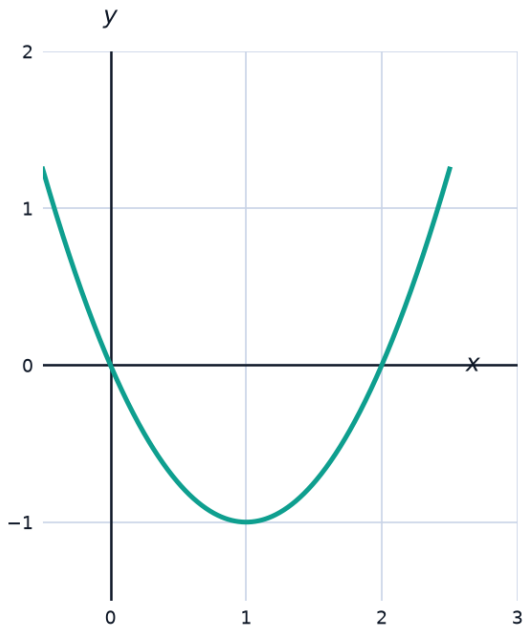


Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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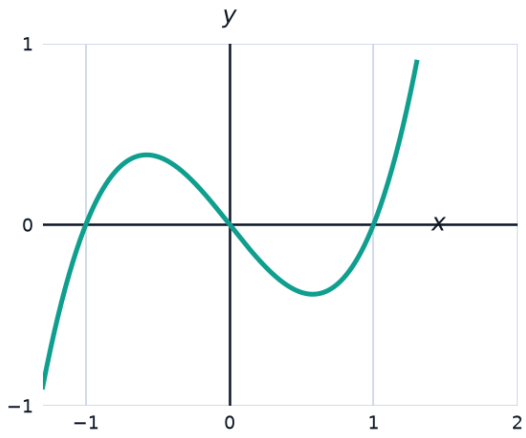
15 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 3.01 The Derivative from First Principles

Find the area of the region bounded by $y = x^2 - 2x$ and the x-axis on $[0, 2]$ (note the curve dips below the axis on this interval).



16 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Find the area between $y = x^3 - x$ and the x-axis on $[-1, 1]$, splitting at the roots $x = -1, 0, 1$ where the sign changes.



17 [Challenging] • Target: Areas involving trigonometric and exponential curves • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

Region bounded by $y = 3 - x^2$ and $y = -1$. Area:.

18 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Volume of region bounded by $x^2/4 + y^2/9 = 1$ rotated about the x-axis:.

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19 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Find the total area enclosed by the cardioid $r = 2(1 - \cos\theta)$.

20 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 6.01 Related Rates

Centroid $\bar{y}$ of the region bounded by $y = x^2$ and $y = 4$:.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area enclosed by $y = x^2$ and $y = 2x - x^2$.

a. 1/3 **b.** 2/3

c. 1/6 **d.** 1

22 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area between $y = x^2$ and $y = 4$.

a. 32/3 **b.** 16/3

c. 8 **d.** 64/3

23 [Foundational] • Target: Vertical Slices (Integration with Respect to x)

Find the area between $y = \sqrt{x}$ and $y = x^2$.

a. 1/3 **b.** 2/3

c. 1/6 **d.** 1/2

24 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Find the area enclosed by $x = y^2$ and $x = y + 2$.

a. 9/2 **b.** 16/3

c. 7/2 **d.** 9

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25 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Why is integrating with respect to y advantageous for curves like $x = y^2 - 4$ and $x = 0$?

a. The right curve minus left curve is a single continuous expression without piecewise splitting

b. Integration with respect to x is impossible

c. It changes the area value

d. The bounds are always 0 and 1

26 [Foundational] • Target: Horizontal Slices (Integration with Respect to y)

Find the area enclosed between $x = 4 - y^2$ and the y-axis.

a. 32/3

b. 16/3

c. 8

d. 16

27 [Foundational] • Target: Crossing Curves and Multiple Regions

Find the total area enclosed between $y = \sin x$ and $y = \cos x$ from $x = 0$ to $x = \pi/2$.

a. $2\sqrt{2} - 2$

b. $\sqrt{2} - 1$

c. 2

d. 1

28 [Foundational] • Target: Crossing Curves and Multiple Regions

Find the area enclosed between $y = x^3 - x$ and $y = 0$ for $x \in [-1, 1]$.

a. 1/2

b. 0

c. 1/4

d. 1

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Crossing Curves and Multiple Regions

Find the area enclosed between $y = x^3$ and $y = 4x$.

a. 8

b. 4

c. 16

d. 0

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30 [Intermediate] • Target: Applications and Boundaries

Find the area of the region bounded by $y = e^x$, $y = e^{-x}$, and $x = 1$.

a. $e + 1/e - 2$

b. $e - 1/e$

c. $e + 1/e$

d. $e - 2$

31 [Intermediate] • Target: Applications and Boundaries

Find the area bounded by $y = \frac{1}{x}$, the x-axis, $x = 1$, and $x = e^2$.

a. 2

b. 1

c. e^2

d. $2e$

32 [Intermediate] • Target: Applications and Boundaries

The Gini index measures income inequality as twice the area between the line of perfect equality $y = x$ and the Lorenz curve $L(x) = x^2$ on $[0, 1]$. Find the Gini index.

a. 1/3

b. 1/6

c. 2/3

d. 1/2

33 [Intermediate] • Target: Area between two curves

Area of the region enclosed by the lemniscate $r^2 = 4\cos 2\theta$:

a. 4

b. 2

c. 8

d. π

34 [Intermediate] • Target: Setting up with respect to y, and total area

Area of region bounded by parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (full astroid):

a. $3\pi/8$

b. $3\pi/4$

c. $\pi/4$

d. $\pi/2$

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35 [Intermediate] • Target: More area problems

Tangent line to parametric curve is horizontal when:

a. $dx/dt = 0$

b. $dy/dt = 0$ and $dx/dte0$

c. $t = 0$

d. $dy/dt = dx/dt$

36 [Intermediate] • Target: Areas requiring case splits

Tangent line to parametric curve is vertical when:

a. $dy/dt = 0$

b. $dx/dt = 0$ and $dy/dte0$

c. $dx/dt = dy/dt$

d. $t = 0$

37 [Intermediate] • Target: Areas involving trigonometric and exponential curves

Pappus's theorem: volume of solid of revolution = $2\pi\bar{x} \cdot A$ where $\bar{x}$ is the centroid of the region and A is the:

a. arc length

b. surface area

c. area of the region

d. cross-sectional area

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Visualizing a region bounded by two curves • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

Surface area of cone with radius r, slant height l:

a. $\pi r^2 l$

b. $\pi r l$

c. $2\pi r l$

d. $\pi r(r + l)$

39 [Challenging] • Target: Area between a curve and a horizontal line • Cross-Lesson: 5.01 The Fundamental Theorem of Calculus

Centroid of semicircular region radius R, $y \geq 0$:

a. $\bar{y} = R/2$

b. $\bar{y} = 4R/(3\pi)$

c. $\bar{y} = 2R/3$

d. $\bar{y} = R/\pi$

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40 [Challenging] • Target: Area between curves in applied contexts • Cross-Lesson: 6.01 Related Rates

Fluid force: pressure at depth d is $\rho g d$. Force on area A:

a. $\rho g A$

b. ρA^2

c. $\rho g d A$

d. $g d$

41 [Challenging] • Target: Vertical Slices (Integration with Respect to x) • Cross-Lesson: 6.02 Optimization and Curve Sketching

Area under one arch of $y = \sin x$ (from 0 to π):

a. 0

b. 2

c. 1

d. π

42 [Challenging] • Target: Horizontal Slices (Integration with Respect to y) • Cross-Lesson: 7.01 Integration by Substitution

Area of astroid $x = \cos^3 t$, $y = \sin^3 t$:

a. $3\pi/4$

b. $3\pi/8$

c. $\pi/2$

d. $3\pi/2$

43 [Challenging] • Target: Crossing Curves and Multiple Regions • Cross-Lesson: 7.02 Integration by Parts

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4

d. 8

44 [Challenging] • Target: Applications and Boundaries • Cross-Lesson: 8.02 Volumes of Revolution

Find the area of the surface formed by revolving $y = \sqrt{x}$, $0 \le x \le 3$, about the x-axis.

a. $\frac{\pi(-1+13\sqrt{13})}{6}$

b. $\frac{\pi(-1+13\sqrt{13})}{12}$

c. $4\sqrt{3}\pi$

d. $\frac{9\pi}{2}$

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45 [Challenging] • Target: Area between two curves • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Find the area of the surface formed by revolving $y = \sqrt{x}$, $0 \le x \le 1$, about the x-axis.

a. $\frac{\pi(-1+5\sqrt{5})}{6}$

b. $\frac{\pi(-1+5\sqrt{5})}{12}$

c. $\frac{4\pi}{3}$

d. $\frac{\pi}{2}$

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